

Weighted Constrained Hue-plane Preserving Camera Characterization

Casper Find Andersen and David Connah

Abstract—Color correction relates device dependent sensor responses (RGB) to device independent color values (XYZ). Here we present a new approach to Hue-plane Preserving Color Correction (HPPCC) using weighted constrained 3×3 matrices. Hue-plane preservation was introduced in [1] in conjunction with an HPPCC method. That method maps using a finite number of local white point preserving 3×3 matrices, each of which operates in a hue-angle delimited subregion of device space defined by the white and two adjacent chromatic training set colors. However, that formulation does not leave room for optimization or continuity beyond C^0 in the transitions between the subregions. To remedy that our new method uses hue-angle specific weighted matrixing: given a device RGB from which a device hue-angle is derived, a corresponding transformation matrix is found as the normalized weighted sum of all precalculated constrained white point and training color preserving matrices. Each weight is calculated as a power function of the minimum difference between the device and the training color hue-angle. The weighting function provides local influence to the matrices that are in close hue-angle proximity to the device color. The power of the function is optimized for global accuracy. We call this Hue-plane Preserving Color Correction by Weighted Constrained Matrixing HPPCC-WCM¹. Experiments performed using different input spectra show that our method consistently improves on both stability and accuracy compared to state of the art methods.

Index Terms—Color Correction, Camera Characterization, Digital Cameras, Geometry, Constrained Least Squares Regression, Color.

I. INTRODUCTION

CAMERA characterisation is an essential element in image processing, since it relates the camera RGB values to colorimetric XYZ values [2], [3] by color correction. This is crucial in any element of digital photography where the images will ultimately be viewed by a human observer, and is a key step in disciplines such as noise reduction, contrast enhancement and gamut mapping, for the same reason. Typically, commercial RGB cameras will have a characterization transform hard-coded into the on-board processing, but bespoke transforms can also be built for lab-based imaging setups [4].

The relation between device RGB and CIE XYZ is not straightforward. There are several reasons for this, the principal one being *filter metamerism*. Filter metamerism is a consequence of the fact that typical RGB cameras do not use

colorimetric sensor filters; i.e. a linear transformation between the CIE color matching functions and the sensor spectral sensitivities does not exist. As a result, the Luther Conditions are not met [5], [6], and multiple color spectra may therefore give rise to identical camera responses, yet have different CIE XYZ values, or vice versa [7]. Although colorimetric filters are technically plausible, they are generally not used due to manufacturing limitations and noise considerations. Therefore optimized rather than exact solutions are usually sought [2], [8], [9].

Typically, optimized transforms are based on linear least squares regression [2], [10], although extensions using non-linear methods have also been proposed. These have the goal of reducing error by increasing degrees of freedom in the optimization, and include polynomial regression [11], [8], [12], [13], [14], neural networks [15], [16], [17] and look-up tables [18]. The regression can also aim to reduce perceptual rather than colorimetric error [19], by minimising errors in a perceptually uniform space. The relative importance of colorimetric or perceptual accuracy depends on the rendering intent, i.e. whether color measurement or pleasing color rendition is the goal [20]. In addition to minimizing average characterisation error, a good solution should conform to several important properties. These include: a) smoothness, and the avoidance of abrupt changes or discontinuities in the mapping; b) predictable and stable mapping for all colours, particularly those with high saturation; c) robustness across different illuminants and reflectance types; d) insensitivity to noise [21]; e) invariance to linear scaling of inputs (exposure independence). Many of these properties are inherent in simple linear transforms, but are either not present, or need to be explicitly imposed in higher order methods. For example, [22] incorporate exposure independence into polynomial transforms, while Finlayson *et al.* [8] show that accurate mapping of the white point can improve perceptual accuracy.

In this paper we introduce a new property, *hue-plane preservation*², and use it to motivate a novel approach to color correction; we call this method “Hueplane Preserving Color Correction by Weighted Constrained Matrixing” (HPPCC-WCM). We demonstrate that as well as reducing color correction error, this approach also naturally incorporates other important properties such as white point preservation, exposure independence, and smoothness, among others.

In Section II we develop the mathematical theory of the relation between sensor responses and spectral reflectances in order to understand how exposure, shading and linearly related

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²An earlier version of the present work can be found in reference [1]

spectral stimuli are represented in sensor space coordinates. In Section III we discuss how these properties work with a representative set of mapping methods; here we specifically define and develop the argument for the incorporation of hue-plane preservation. In Section IV our new approach to hue-plane preserving color correction is presented. Results of a comparison between the new method and the choice of existing state-of-the-art methods are presented in Section VI and discussed in VII. We conclude in Section VIII.

II. BACKGROUND

The basic method for computing an optimised solution is well established [23] [4]. A color calibration chart with a number of patches of known XYZ tristimulus values (and sometimes known reflectance spectra) is imaged with a digital camera, taking care to ensure uniform illumination on the target and the removal of stray light, and an exposure is chosen to minimize noise. The RGB responses for each patch are recorded in the camera and the parameters of a function are estimated to map RGB values to estimated XYZ values.

A. Color formation

The standard model of the image formation process is given in Equations 1 and 2. Sensor responses $\mathbf{s}$ to a spectral power distribution $P(\lambda)$ are given by:

$$\mathbf{s} = \int_{\lambda} P(\lambda) \sigma(\lambda) d\lambda \quad (1)$$

where λ is wavelength, $\sigma(\lambda)$ are the spectral filter functions of the sensors and:

$$P(\lambda) = I(\lambda) \rho(\lambda) \quad (2)$$

where $I(\lambda)$ is the illuminant spectral power distribution and $\rho(\lambda)$ is a surface. In case the sensor is a camera, we denote the corresponding sensor responses $\mathbf{c}$. Specifically, in the usual 3 filter case which is relevant here $\mathbf{c} = [R, G, B]^T$. In case the sensor is the human standard observer color matching functions, we denote the corresponding responses $\mathbf{x} = [X, Y, Z]^T$. In Equation 1 we assume that $\sigma(\lambda) = [w_1\sigma_1(\lambda), w_2\sigma_2(\lambda), w_3\sigma_3(\lambda)]$, where $\sigma_i(\lambda)$ is the i^{th} sensitivity function, $i = (1, 2, 3)$ and w_i represents pre-calculated white-point balancing factors in either RGB or XYZ-space.

B. Characterization models and optimization

A characterization model seeks to map device dependent response vectors $\mathbf{c}$, to approximate device independent tristimulus values $\hat{\mathbf{x}}$ with a transformation function $f(\cdot)$:

$$\hat{\mathbf{x}} = f(\mathbf{c}) \quad (3)$$

Although we refer to XYZ values throughout, any set of tristimulus values can be interchanged.

Generally, and most importantly, $f(\cdot)$ must be error minimising and is found by:

$$\min_f \sum_{i=1}^N \|u(\mathbf{D}\mathbf{x}_i) - u(\mathbf{D}f(\mathbf{c}_i))\|_{\nu} \quad (4)$$

where $\mathbf{x}_i$ and $\mathbf{c}_i$ are the observer and camera responses to the i^{th} of N characterization target patches, $u(\cdot)$ is a color space transformation, and $\mathbf{D}$ is an $N \times N$ diagonal matrix containing normalised weights for each color $\mathbf{c}_i$ that can be employed to bias the minimisation towards some colors more than others. ν represents some image norm. Most commonly $u(\cdot)$ is the identity function, $\mathbf{D}$ is the identity matrix and $\nu = 2$, for Euclidean distances. Equation 4 then becomes:

$$\min_f \sum_{i=1}^N \|\mathbf{x}_i - f(\mathbf{c}_i)\|_2 \quad (5)$$

In the case where $f(\cdot)$ is linear:

$$f(\mathbf{c}) = \mathbf{M}_{lsq} \mathbf{c} \quad (6)$$

where $\mathbf{M}_{lsq}$ is a 3×3 matrix. This is known as the standard unconstrained linear least squares solution to Equation 3.

However, $u(\cdot)$ can also be a mapping to a more perceptually uniform space, such as CIELAB (ISO 11664-4:2008(E)/CIE S 014-4/E:2007) to ensure the perceptual accuracy of the final characterisation [19]. Furthermore, the optimisation in Equation 4 can be adapted to incorporate desirable properties, such as white point balancing, through appropriate constraints.

C. Exposure and shading

The inherent linearity of both the standard observer and the RGB sensor responses to light means that scaling by a factor k , caused by a change in exposure or irradiance, will result in an equal change in both responses. The color signal $P(\lambda, k)$ after scaling by k can be represented as:

$$P(\lambda, k) = kI(\lambda) \rho(\lambda) \quad (7)$$

Inserting 7 into Equation 1 yields:

$$\mathbf{s}(k) = \int_{\lambda} \rho(\lambda, k) \sigma(\lambda) d\lambda \quad (8)$$

where the response to the scaled reflectance is $\mathbf{s}(k)$. This tells us that the camera response:

$$\mathbf{c}(k) = k\mathbf{c} \quad (9)$$

is scaled in exactly the same way as the observer response:

$$\mathbf{x}(k) = k\mathbf{x} \quad (10)$$

We argue that this property of equal scaling, which we refer to as *exposure or shading independence*, should be maintained by characterization transforms so that:

$$f(k\mathbf{c}) = kf(\mathbf{c}) \Rightarrow k\hat{\mathbf{x}} = f(k\mathbf{c}) \quad (11)$$

and therefore any scaled version of a color automatically will be mapped to the appropriately scaled version of the estimated color. Enforcing this constraint provides an inherent colorimetric accuracy and ensures that the same characterisation is valid if, for example, the scene lighting level changes.

D. Interest color and white point preservation

There are many situations where it is desirable to map particular colors exactly to their correct XYZ values; for example in cultural heritage, in medical applications, for specific brands, or objects with colors of high importance. Letting $u(\cdot)$ be the identity function henceforth, this is expressed mathematically as solving Equation 4 with the constraint that:

$$f(\mathbf{c}_k) \equiv \mathbf{x}_k \quad (12)$$

i.e. a k^{th} color $\mathbf{c}_k$ must be mapped with no error. We will refer to this property as *interest color preservation*.

A key property preserved by a number of transforms is the imaging “white-point”, which is a specific version of the preservation of interest colors. This can be expressed mathematically as:

$$f(\mathbf{c}_W) \equiv \mathbf{x}_W \quad (13)$$

where we note that $\mathbf{x}_W$ is the exact white-point, rather than $\hat{\mathbf{x}}_W$ which is the approximation to the white point. This property is referred to as *white point preservation* and it has the perceptually important consequence that color casts, especially in neutral colours, are avoided.

We also note here that preserving the white-point is not the same as white-point balancing. In the camera characterization pipeline a white point balancing step should be applied independently and before optimisation [4]. Additionally, a separate characterization transform should be built for each illuminant that is encountered, and typically commercial cameras will have multiple characterization functions to apply under different illumination conditions.

E. Hueplane preservation

Using Shafer’s dichromatic reflection model [33] (also known as the Phong model in computer graphics [24], [25]) let the reflectance from an object be a linear combination of two spectral reflectances; diffuse reflectance from a surface pigment and a neutral specular reflectance so that:

$$\rho(\lambda, k_D, k_S) = (k_D \rho_D(\lambda) + k_S) \quad (14)$$

where $\rho(\lambda, k_D, k_S)$ is the surface spectral reflectance function expressed as a linear combination of specular reflectance (weighted by the amount k_S) and diffuse pigment reflectance $\rho_D(\lambda)$ (weighted by the amount k_D). The integrated camera response can now be written:

$$\mathbf{c}(k_D, k_S) = k_D \mathbf{c}_D + k_S \mathbf{c}_S \quad (15)$$

and the response of the observer as:

$$\mathbf{x}(k_D, k_S) = k_D \mathbf{x}_D + k_S \mathbf{x}_S \quad (16)$$

where $\mathbf{c}_D$ is the 3-element camera response to the diffuse reflectance component and $\mathbf{c}_S$ to the specular component (or light source), $\mathbf{x}_D$ is the observer response to the diffuse component and $\mathbf{x}_S$ to the specular component. Assuming that the responses to the illuminant are scaled by a factor k_W , bringing the white point within exposure range, we can

replace $k_S \mathbf{c}_S$ with $k_W \mathbf{c}_W$ and $k_S \mathbf{x}_S$ with $k_W \mathbf{x}_W$ and thus $\rho(\lambda, k_D, k_S)$ with $\rho(\lambda, k_D, k_W)$. It is clear from Equations 15 and 16 that a) both the camera and observer responses to the stimulus $\rho(\lambda, k_D, k_W)$ are the weighted combination of the separate responses to the diffuse and specular components of the color, and b) that the weights k_D and k_W are the same both for camera and observer. By varying k_D and k_W in positive numbers a halfplane in RGB space corresponding to a halfplane in XYZ space is described. We denote these planes *hue-planes*. In general terms:

$$k_D f(\mathbf{c}_D) + k_W f(\mathbf{c}_W) = f(k_D \mathbf{c}_D + k_W \mathbf{c}_W) \quad (17)$$

and using Equation 3:

$$k_D \hat{\mathbf{x}}_D + k_W \hat{\mathbf{x}}_W = k_D f(\mathbf{c}_D) + k_W f(\mathbf{c}_W) \quad (18)$$

A useful way to understand the position and geometry of a hueplane is to compute chromaticity coordinates (r, g) for camera color $\mathbf{c}$, given by:

$$r = R/(R + G + B), \quad g = G/(R + G + B) \quad (19)$$

Scaling the color values (R, G, B) by a constant k and inserting this into 19:

$$r = kR/(kR + kG + kB), \quad g = kG/(kR + kG + kB) \quad (20)$$

shows that chromaticity planes are invariant to changes in exposure and shading. Inserting the camera white point $\mathbf{c}_W = (R_W, G_W, B_W)$ in Equation 19 gives the neutral point $(r_W, g_W) = (\frac{1}{3}, \frac{1}{3})$. The chromaticity point (r, g) and the neutral point (r_W, g_W) will define a half line emanating from the neutral point, extending to the color’s dominant wavelength [3]. This line in turn defines an angle ϕ_c by:

$$\phi_c = \text{sign}\left(\frac{v}{C} \arccos\left(\frac{u}{C}\right)\right) + \frac{1}{2}(1 - \text{sign}\left(\frac{v}{C}\right)) \quad (21)$$

where $\phi_c \in [0; 2\pi]$ and the chroma C is defined as:

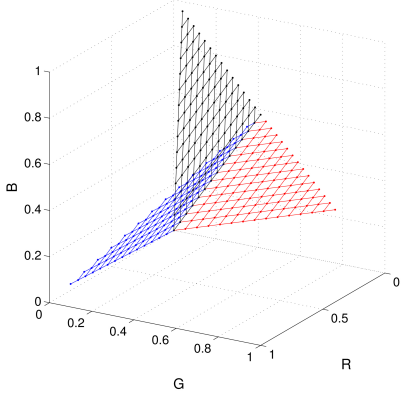
$$C = \sqrt{(u^2 + v^2)} \quad (22)$$

and:

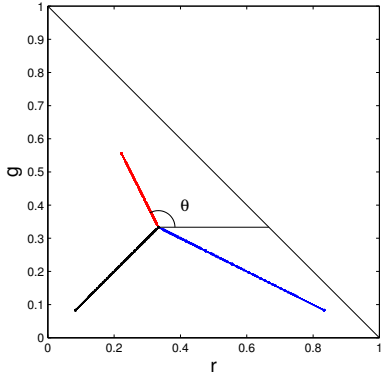
$$u = (r - r_W), \quad v = (g - g_W) \quad (23)$$

The angle ϕ is referred to as a *hue-angle*. Combining the linear relations in 18 with white point preservation in Equation 13, we will refer to this property of the mapping defined by Equation 3 as *hueplane preservation*. White point preservation, exposure and shading independence are necessary but not sufficient prerequisites for hue-plane preservation, as the root polynomial methods possess these properties but hue-planes are generally curved by those mappings.

In Figure 1 a) we show a graphical representation of three hue planes represented in the RGB color space. In Figure 1 b) we show the hue-planes in the (r, g) chromaticity space. Here the chromaticity lines corresponding to each plane are clearly visible, with the hue-angle represented visually as it is defined above.



(a) 3D hue planes in RGB space.



(b) 2D hue planes in (r, g) with hue-angle

Fig. 1. A visual representation of three hue planes in a) the camera's native RGB space and b) (r, g) chromaticity space.

The importance of hue-plane preservation can be seen in real scenes with 3-dimensional objects having uniform, or near-uniform, diffuse reflectance (i.e. conforming to the Shafer, or Phong model). In this case, if the diffuse color is corrected without error, or low error, all other colors from that object are automatically mapped accordingly with no or low error. It is important to note that a hue-plane is correlated to perceptual hue by dominant wavelength. This means that a hue-plane does not necessarily contain perceptually uniform hue over its surface, just as an object with Phong modelled reflection is not necessarily perceived to have a uniform hue.

III. EXISTING COLOR CORRECTION METHODS

A. Linear regression

Adding a white-point preserving constraint defined in Equation 13 to the linear solution in Equation 5 we obtain:

$$\hat{\mathbf{x}} = \mathbf{M}_l \mathbf{c} \text{ subject to } \mathbf{M}_l \mathbf{c}_W \equiv \mathbf{x}_W \quad (24)$$

We will refer to this as the linear method (LIN). Applying LIN to hue-plane preservation defined in Equation 17:

$$k_D \mathbf{M}_l \mathbf{c}_D + k_W \mathbf{M}_l \mathbf{c}_W = \mathbf{M}_l (k_D \mathbf{c}_D + k_W \mathbf{c}_W) \quad (25)$$

reveals that $\mathbf{M}_l$ is indeed hue-plane preserving. The drawback of the LIN method is the comparatively low accuracy of the

estimations, mainly caused by the low number of degrees of freedom in the matrix, having only 9 coefficients of which 3 are occupied by the white point preservation, which means that only 2 hueplanes can be mapped exactly.

B. Polynomial regression

In order to enhance the accuracy found in the linear approach a higher order function $f(\cdot)$ can be employed while retaining white point preservation. The typical polynomial approach to color correction maps a polynomial expansion of the device RGB values to estimated XYZ. A standard 2^{nd} order polynomial expansion [22] of $\mathbf{c}$ becomes $\mathbf{c}^{(2)}$:

$$\mathbf{c}^{(2)} = [R, G, B, R^2, G^2, B^2, RG, RB, BG]^T \quad (26)$$

and a 3^{rd} order polynomial expansion of $\mathbf{c}$ adds third order elements to $\mathbf{c}^{(2)}$:

$$\mathbf{c}^{(3)} = [\mathbf{c}^{(2)T}, R^3, R^2G, R^2B, G^3, G^2R, G^2B, B^3, B^2R, B^2G, RGB]^T \quad (27)$$

Either way, the white point preserving polynomial mapping function becomes:

$$\hat{\mathbf{x}} = \mathbf{M}_{pol} \mathbf{c}^{(q)} \text{ subject to } \mathbf{M}_{pol} \mathbf{c}_W^{(q)} \equiv \mathbf{x}_W \quad (28)$$

where $\mathbf{M}_{pol}$ is a 3×9 matrix if $q = 2$ and a 3×19 matrix if $q = 3$. We will refer to these polynomial mappings as POL2 and POL3 in our experiments.

Applying POL2 and POL3 to the color signal $\mathbf{c}$ in Equation 17, it is clear that the higher order elements will prevent hue-plane preservation, as well as exposure and shading independence. This is also true for neural-network approaches, or any other non-linear models where exposure independence and linearity are not explicitly factored in. As such these methods are only useful for a fixed imaging set up, e.g. when imaging objects in the laboratory under a fixed light source.

C. Root polynomial regression

Finlayson *et al.* [22] have proposed expansions based on root polynomials that conform to exposure independence, with promising results in colorimetric accuracy.

A 2^{nd} order root polynomial will have the 6 components [22]:

$$\mathbf{c}^{(1/2)} = [R \ G \ B \ \sqrt[3]{RG} \ \sqrt[3]{RB} \ \sqrt[3]{BG}]^T \quad (29)$$

while a 3^{rd} order root polynomial will have the 13 components:

$$\mathbf{c}^{(1/3)} = [\mathbf{c}^{(1/2)T} \ \sqrt[3]{R^2B} \ \sqrt[3]{R^2G} \ \sqrt[3]{G^2R} \ \sqrt[3]{G^2B} \ \sqrt[3]{B^2R} \ \sqrt[3]{B^2G} \ \sqrt[3]{RGB}]^T. \quad (30)$$

Inserting $\mathbf{c}^{(1/2)}$ or $\mathbf{c}^{(1/3)}$ into Equation 5 gives:

$$\hat{\mathbf{x}} = \mathbf{M}_{rpol}\mathbf{c}^{(1/q)} \text{ subject to } \mathbf{M}_{rpol}\mathbf{c}_W^{(1/q)} \equiv \mathbf{x}_W \quad (31)$$

We will refer to the root polynomial mapping methods as RPOL2 and RPOL3. Applying RPOL2 and RPOL3 to the color signal $\mathbf{c}$ in Equation 17 reveals that by setting either $k_D = 0$ or $k_W = 0$ the function $f(\cdot)$ is exposure and shading independent, but that the higher order terms exclude hue-plane preservation.

We expect that the inability to preserve hue-planes in these higher order methods, will have an impact on very saturated colors. In particular we expect that the lower the order, the less curvature and the less accuracy (1^{st} order polynomial is identical to the low accuracy 3×3 matrix), whereas higher orders may fit extreme values poorly [26]. We demonstrate this in Section VII-C.

D. Enhanced hue-plane preserving matrixing

In order to increase the number of degrees of freedom in the mapping function $f(\cdot)$, but retain hue-plane preservation, Find Andersen and Hardeberg [1] constructed a piecewise linear method using multiple white point preserving 3×3 matrices. To define the matrices, they impose interest point preservation on the training colors. So for any pair of characterisation patches in a j^{th} interval $\mathbf{c}_j$ and $\mathbf{c}_{j+1}$, where j indicates a rank in terms of increasing camera hue, a hue interval delimited subregion of camera color space is created between ϕ_j and ϕ_{j+1} . A matrix $\mathbf{M}_j$ in that interval is found such that:

$$\mathbf{M}_j\mathbf{c}_W \equiv \mathbf{x}_W \text{ and } \mathbf{M}_j\mathbf{c}_j \equiv \mathbf{x}_j \text{ and } \mathbf{M}_j\mathbf{c}_{j+1} \equiv \mathbf{x}_{j+1} \quad (32)$$

where there are as many matrices as training colors. For a generic device color $\mathbf{c}$ the appropriate color correction matrix $\mathbf{M}_k$ is then chosen such that the hue of $\mathbf{c}$ falls between the hue values of ϕ_k and ϕ_{k+1} . The correction result will still only be approximate for colors that do not belong to the characterization target. They term this approach ‘‘hue-plane preserving color correction’’ (HPPCC).

Increasing the number of matrices in the color correction allows significantly decreased error compared to a single 3×3 matrix. But this method has drawbacks in that smoothness beyond C^0 is not ensured and the method doesn’t leave room for optimization except by manually choosing characterization patches. This choice of color patches makes the estimation error sensitive in areas of the color space, defined by the data set, with a high level of metamerism.

IV. THE WEIGHTED CONSTRAINED MATRIXING METHOD

We adress the above issues directly in our HPPCC-WCM method motivated by the drive to ensure that the characterization is hueplane preserving and crucially, error minimising.

The present work extends the initial HPPCC idea by changing the mapping from piecewise linear to differentially continuous (C^∞). This entails preserving smoothness in the transformation between physically smooth color transitions recorded by the camera and equally smooth color transitions recorded by the observer. The only place in color space in which this method may be reduced to C^0 continuity is in

passing the neutral axis. But since neither hue nor saturation are perceptually present there, no abrupt perceptual color shifts can occur there. Like the standard polynomial method presented in the previous section, HPPCC-WCM employs only one user controlled parameter that directly relates to the order of differentiability limited by the number of training colors.

The estimated color $\hat{\mathbf{x}}$ corresponding to $\mathbf{c}$ is, for the HPPCC-WCM method, given by:

$$\hat{\mathbf{x}} = \mathbf{M}(\phi_c, p)\mathbf{c} \quad (33)$$

Given the device color $\mathbf{c}$ from which a device hue-angle ϕ_c is derived by Equation 21, the corresponding transformation matrix $\mathbf{M}(\phi_c, p)$ is found as the normalized power function weighted sum of all precalculated matrices $\mathbf{M}_i$ for $i \in [1 : N]$, each corresponding to a training color $\mathbf{c}_i$. The parameter p refers to the order of the weighting power function: To obtain $\mathbf{M}(\phi_c, p)$ the N constrained 3×3 matrices $\mathbf{M}_i$ for each training color $\mathbf{c}_i$, are thus found by equation constrained least squares regression:

$$\min_{\mathbf{M}_i} \sum_{i=1}^N \|\mathbf{M}_i\mathbf{c}_i - \mathbf{x}_i\| \quad (34)$$

subject to the constraints of white point preservation and interest color preservation of training color $\mathbf{c}_i$ as defined in Equations 12 and 13:

$$\mathbf{M}_i\mathbf{c}_W = \mathbf{x}_W \text{ and } \mathbf{M}_i\mathbf{c}_i = \mathbf{x}_i \quad (35)$$

The N matrices $\mathbf{M}_i$ are weighted as a function of hue-angles ϕ_c and each of the N training color hue-angles ϕ_i . The weighting function $w(\cdot)$ is defined as a power function of order p of hue-angle differences:

$$w(\phi_i) = (\pi - \Delta\phi_i)^p \quad (36)$$

where the i^{th} hue-angle difference $\Delta\phi_i = \min(|\phi_c - \phi_i|, 2\pi - |\phi_c - \phi_i|)$. Finally, $\mathbf{M}(\phi_c, p)$ in Equation 33 is calculated as the normalized weighted sum:

$$\mathbf{M}(\phi_c, p) = \frac{1}{\sigma_w} \sum_{i=1}^N w(\phi_i)\mathbf{M}_i \quad (37)$$

where:

$$\sigma_w = \sum_{i=1}^N w(\phi_i) \quad (38)$$

Letting p go towards infinity for $\phi = \phi_i$ will force $\mathbf{M}(\phi, p)$ to go towards $\mathbf{M}_i$ and generally make matrix influence become very local to matrices pertaining to hues that are extremely close to the hue of the color to be mapped. Letting $p = 0$ will put equal weight on all $\mathbf{M}_i$ and thus approach the solution calculated by Equation 13. By varying p the estimation in Equation 33 can be optimized for accuracy. Using Equation 17 demonstrates that HPPCC-WCM indeed is hue-plane preserving.

V. EQUATION CONSTRAINED LEAST SQUARES REGRESSION METHOD

In order to obtain the linear least squares solution to Equations 5, 6, 28, 31 and 34, subjected to particular constraints, we begin by expressing the constraints as a set of linear equations:

$$\mathbf{E}\vec{m} = \vec{x}_e \quad (39)$$

where $\vec{m}_{i+3(j-1)} = m_{j,i}$, for $i \in [1:3]$ and $j \in [1:3]$, is a 9×1 vector representation of $\mathbf{M}$. $\mathbf{E}$ is a $3k \times 9$ matrix containing the constraints corresponding to k interest color preservations, where $\mathbf{E} = \mathbf{0}$ except for $e_{j+3(k-1),3(j-1)+i} = c_i^k$ and $\vec{x}_{i+3(k-1)} = x_i^k$, in which case $\mathbf{c}^k$ and $\mathbf{x}_e^k$ are the camera and colorimetric values for a k^{th} constraint.

These constraints are imposed on the linear optimization of the color mapping:

$$\mathbf{P}\vec{m} = \vec{x} \quad (40)$$

where $\mathbf{P}$ is a $3N \times 9$ matrix where $\mathbf{P} = \mathbf{0}$ except for $p_{j+3(k-1),3(j-1)+i} = c_i^k$ and $\vec{x}_{i+3(k-1)} = x_i^k$, and again $\mathbf{c}^k$ and $\mathbf{x}^k$ are the camera and colorimetric values for a k^{th} training color.

To solve this system we use the method in [27], which is an equation constrained linear least squares solving method based on Lagrange multipliers λ . That is, we minimize:

$$\min_{\vec{m}=\vec{x}_e} \|\mathbf{P}\vec{m} - \vec{x}\|_2 \quad (41)$$

by using the unique solution to:

$$\begin{pmatrix} \hat{m} \\ \hat{\lambda} \end{pmatrix} = \mathbf{B}^{-1} \begin{pmatrix} \mathbf{P}^T \vec{x} \\ \vec{x}_e \end{pmatrix} \quad (42)$$

where:

$$\mathbf{B} = \begin{bmatrix} \mathbf{P}^T \mathbf{P} & \mathbf{E}^T \\ \mathbf{E} & \mathbf{0} \end{bmatrix} \quad (43)$$

VI. RESULTS

A. Optimising p

The method uses a single parameter, p , which defines the weighting given to each training matrix. Given that there is no analytical solution for p , it must be optimised numerically for a set of training patches. Figure 2 shows average XYZ error for different values of p using a leave-one-out (LOO) testing regime for two different characterisation targets; the Macbeth ColorChecker (MbCC) with 24 patches, and the Macbeth ColorChecker DC (MbDC) with 238 patches.

Given that colors close to the neutral axis are already mapped accurately by white point preservation in our method and that only one shade (preferably the one with highest reflectance) of each training color is needed to define each hueplane, we only use chromatic patches to generate training matrices. For the MbCC chart we use the 18 non-neutral patches, while for the MbDC chart we use a chroma threshold (see Eq. 22) to remove all non-neutral patches; a value of 0.05 was found to be most effective, leaving 127 matrices. Training error was then evaluated on all patches, chromatic and neutral. When the color to be “left out” is one of the chromatic training patches, only the remaining training patches (17 and

126 for the MbCC and MbDC respectively) are used to train the method. The results show that the optimal value of p falls at different values for the different datasets.

To find an optimal value for p , we use non-linear numerical optimisation to minimise least-squares XYZ error for a given imaging set up and set of training data. In a practical setting, this value would only need to be computed once for a fixed characterisation mapping; here we minimise LOO error to do this. Table I shows optimal p values averaged for 31 different cameras, 4 illuminants, and two characterisation charts. As above, there is large variability between the two characterisation charts, but for each chart the variability across illuminant and cameras is relatively small. Indeed, if the mean value of p is used for all cameras rather than optimising for each, there is a drop in performance of less than $0.05 \Delta E$. In later results, to simplify the experimental protocol, a single p value is used over all cameras for a given illuminant.

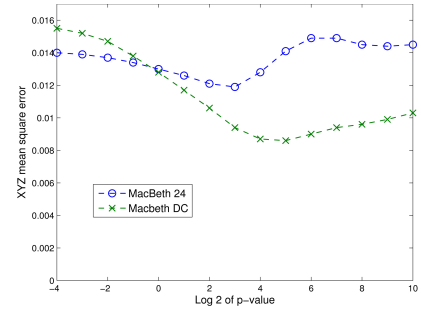


Fig. 2. XYZ estimation error plotted as a function of p for two different datasets using a leave-one-out training regime.

TABLE I
OPTIMISED VALUES FOR p

		MbCC			MbDC		
	Mean	Min	Max	Mean	Min	Max	
D65	10.7	5.1	15.5	32.4	22.4	85.6	
A	10.8	2.5	21.4	64.3	40.6	102.0	
F11	4.3	2.3	11.2	38.8	27.7	60.4	
D50	12.7	7.1	17.5	33.0	23.6	62.4	

Although the non-linear training of HPPCC-WCM is computationally most complex, once the characterization transform has been built, the LIN, RPOL and HPPCC-WCM are all $O(N)$ since the required number of computational operations are proportional to the number of pixels. The LIN method is the least complex, requiring a 3×3 multiplication per pixel. The polynomial methods require the expansion of each RGB into higher order coefficients prior to matrix multiplication. In the WCM method the hue must first be computed using Equation 21 per pixel, by which a weighted 3×3 matrix can be calculated and applied. However, in practice a $K_h \times 9$ look up table could be built from K_h discrete hue values to pre-computed transform matrices, since the derivation of the transform can be performed off-line.

B. Training and Testing

We test our methods on a mixture of real and synthesized data. We primarily use synthesized data: a) because the responsiveness of CCD sensitivity is well understood [30] and hence Equation 1 provides an accurate model of real data, and b) because it allows us to test the method on a large number of different cameras and spectral datasets. We use the real data to validate these results.

To keep our simulations realistic, and practically relevant, we use recognised calibration charts as training data; here we use the (*MbCC*) and (*MbDC*) charts, to evaluate performance on different sized training sets. To choose optimal p , we minimise *LOO* error separately for each.

As testing data in our simulations we use carefully chosen reflectance datasets to represent different types of possible spectra: 1) Munsell reflectances [28] as an example of man-made spectra; 2) the Esser TE221 test chart as an example of challenging man-made spectra (the Esser chart contains some very high chroma colors); 3) natural spectra from Foster *et al.* [29]; 4) colors generated on the same hue-plane as training patches, using the Phong shading model. For this *hue-planes* set, reflectances are sampled at equally spaced intervals on a triangular grid in reflectance space, where the triangle is anchored by the perfect reflecting diffuser, black point, and the reflectance of the color in question. This simulates the situation where uniformly colored, diffusely and specularly reflecting, 3D objects are photographed under the imaging illuminant.

The data are synthesized according to Equation 1. In our experiments RGB values are generated for 31 different cameras, 28 from [31] and 3 from [32]. Of the many existing illuminants, we have chosen three representative of indoor, outdoor and color management, D65 (a CIE standard illuminant representative of daylight), A (Incandescent/tungsten physical source) and F11 (Phillips TL84, representative of indoor fluorescent illumination). In addition, we test under illuminant D50, which is a second daylight simulator that is often used in ICC Color Management.

In our practical imaging experiments we use the Phase One 80MP camera to capture the (2014 by X-Rite) MacBeth 24 patch ColorChecker and 140 patch Gretag Macbeth SG ColorChecker under tungsten illumination in a controlled laboratory setting³.

Tables II and IV show results for *LOO* error evaluation on the synthesised data using the MbCC and MbDC charts respectively. Results show the mean, median and max CIELab error over 31 cameras, and under three illuminants. The methods compared are: linear white-point preserving matrix (LIN) [8], polynomial (POL2, POL3) [12], root polynomial (RPOL2, RPOL3) [22] and HPPCC-WCM, where all methods include white point preservation. Tables III and V show results for a realistic training / testing situation, using the MbCC and MbDC charts respectively as training data, and 4 test reflectance sets. For each table the best performing method is highlighted in bold for each illuminant and error metric.

Figure 3 demonstrates the effect of the different methods on hue-planes. In Figure 3 a) and b) XYZ and xy values are

shown respectively for RGB planes passed through the RGB to XYZ conversion. In Figure 3 b) true xy chromaticities for a hue-plane derived from a chromatic Esser patch are plotted against estimations from four different approaches.

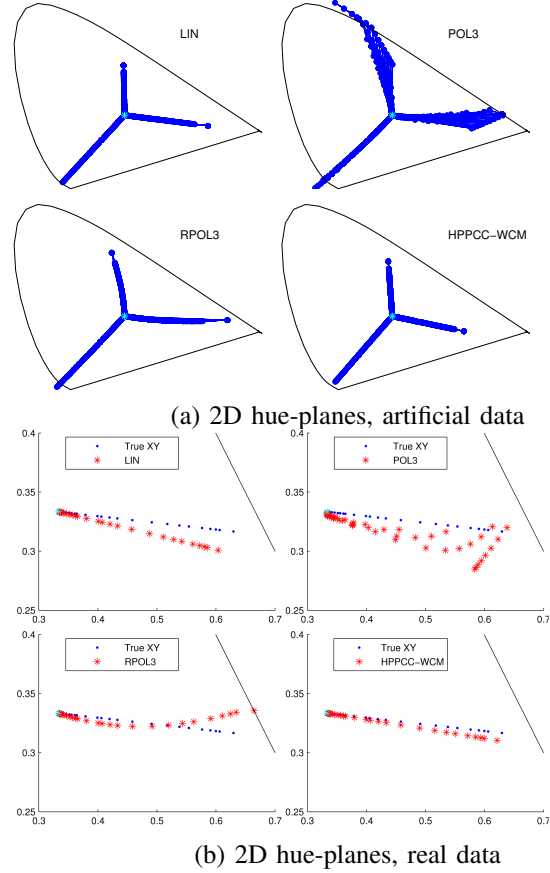


Fig. 3. Visualisations of hue-planes plotted on a CIE xy chromaticity diagram after mapping to XYZ: a) shows the effect of each transform on constant RGB hue-planes (shown in Figure 1), b) shows a hue-plane generated from a chromatic patch taken from the Esser data (see text for details). Curvature imposed by the POL3 and RPOL3 methods is clear in each case.

TABLE VII
CHARACTERISATION PERFORMANCE ON REAL DATA. TRAINING DATA: MACBETH 24 PATCHES COLORCHECKER. TESTING DATA: 140 PATCHES OF THE MACBETH SG COLORCHECKER. THE HPPCC-WCM METHOD USES A p VALUE OF 8.6.

Method	Mean	Median	Max
LIN	5.2	3.2	40.4
POL2	4.5	3.5	31.8
RPOL2	5.2	2.9	40.6
HPPCC-WCM	4.4	2.9	40.6

Table VI shows the effect of adding noise to sensor outputs on test data (noise in training data is generally minimal due to averaging responses over training patches). Here we add both dark noise and shot noise; dark noise is modelled as zero mean Gaussian noise distributed $N(0, 0.01)$, and shot noise as an approximation to Poisson noise using Gaussian noise whose standard deviation is scaled by the sensor intensity, i.e. for a sensor response c_i the noise has distribution $N(0, 0.02c_i)$.

³Data supplied by PhaseOne, Denmark and used with permission

TABLE II

CIELAB ΔE ERROR (MEAN, MEDIAN AND MAX) USING LEAVE-ONE-OUT TESTING PARADIGM FOR EACH METHOD, WITH RESULTS AVERAGED OVER 31 CAMERAS; REFLECTANCE SET: MACBETH 24 PATCH COLORCHECKER, ALL ILLUMINANTS. THE HPPCC-WCM USES A p VALUE OF 8.6.

		D65			A			F11			D50	
Method	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
LIN	1.9	1.4	29.3	2.3	1.5	28.8	1.5	1.1	18.8	2.0	1.4	28.6
POL2	2.6	1.6	23.3	3.1	1.7	39.7	2.3	1.2	23.0	2.7	1.6	25.8
POL3	3.2	1.7	59.6	4.2	2.4	67.0	2.7	1.5	35.8	3.1	1.9	51.4
RPOL2	1.9	1.1	20.2	2.4	1.3	29.5	1.8	1.1	17.5	2.0	1.2	20.5
RPOL3	3.9	2.0	82.6	5.3	2.8	91.2	4.9	1.7	68.6	4.2	1.9	81.6
HPPCC-WCM	1.7	1.0	23.2	2.2	1.3	40.7	1.4	0.8	13.8	1.7	1.0	25.7

TABLE III

CIELAB ΔE ERROR (MEAN MEDIAN AND MAX) FOR A RANGE OF TEST SETS AFTER TRAINING ON MACBETH 24 PATCH COLORCHECKER. RESULTS ARE AVERAGED OVER 31 CAMERAS, AND PRESENTED FOR THREE DIFFERENT IMAGING ILLUMINANTS. THE HPPCC-WCM USES A p VALUE OF 8.6 FOR EACH CASE.

			D65			A			F11			D50	
Training	Method	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
MbCC	LIN	1.7	1.2	22.1	2.1	1.3	22.9	1.4	0.9	15.5	1.7	1.2	21.4
(N=24)	POL2	1.5	1.0	16.8	1.7	1.2	15.5	1.3	0.9	10.1	1.5	1.0	17.5
	POL3	0.3	0.2	4.2	0.4	0.2	4.8	0.2	0.1	3.3	0.3	0.2	4.6
	RPOL2	1.3	0.8	11.3	1.6	0.9	16.0	1.2	0.7	10.7	1.3	0.8	12.0
	RPOL3	0.7	0.5	7.4	0.9	0.6	9.6	0.8	0.7	4.9	0.8	0.5	8.6
	HPPCC-WCM	1.0	0.6	16.0	1.3	0.7	22.8	0.9	0.5	9.8	1.1	0.5	17.1
Testing			D65			A			F11			D50	
Esser	LIN	2.8	1.6	47.6	3.0	1.8	45.9	2.4	1.4	39.1	2.8	1.6	47.2
(N = 264)	POL2	2.6	1.4	38.6	3.2	1.7	46.0	2.4	1.2	56.2	2.7	1.5	39.0
	POL3	6.6	2.4	305.7	7.2	2.5	266.9	5.7	2.1	167.7	6.8	2.4	295.2
	RPOL2	1.9	1.1	35.0	2.4	1.2	53.2	2.1	0.9	37.3	2.0	1.1	35.2
	RPOL3	4.2	1.4	238.8	5.8	1.9	193.8	3.6	1.4	218.5	4.5	1.5	238.9
	HPPCC-WCM	2.0	1.1	39.0	2.4	1.4	45.8	1.7	1.0	39.1	2.1	1.2	40.5
Munsell	LIN	1.5	0.9	45.9	1.7	1.0	48.8	1.4	0.8	30.1	1.6	0.9	46.4
(N = 1269)	POL2	1.3	0.9	30.3	1.5	0.9	63.1	1.2	0.7	16.1	1.3	0.9	42.0
	POL3	3.1	1.5	231.5	3.3	1.5	194.7	3.1	1.5	231.5	3.2	1.5	196.9
	RPOL2	1.1	0.7	40.9	1.3	0.8	49.1	1.1	0.6	15.2	1.1	0.7	46.7
	RPOL3	1.4	0.8	81.2	1.8	0.9	51.1	1.4	0.8	81.2	1.4	0.8	74.9
	HPPCC-WCM	1.3	0.7	46.8	1.6	0.8	61.9	1.2	0.6	29.1	1.3	0.7	47.2
Natural	LIN	2.1	1.3	49.7	2.4	1.3	53.2	2.2	1.2	40.7	2.1	1.2	50.0
(N = 4871)	POL2	2.1	1.4	104.3	2.3	1.3	151.7	1.9	1.2	53.5	2.0	1.3	106.4
	POL3	3.7	1.5	1010.5	4.0	1.4	480.0	4.1	1.2	895.1	3.8	1.4	1014.0
	RPOL2	2.1	1.3	43.0	2.4	1.4	59.1	1.9	1.2	27.2	2.0	1.3	46.7
	RPOL3	2.4	1.5	173.1	3.1	1.9	103.7	2.0	1.4	64.8	2.3	1.5	150.7
	HPPCC-WCM	2.1	1.3	49.7	2.4	1.3	76.1	1.9	1.1	40.7	2.1	1.2	50.0
Hue planes	LIN	0.8	0.4	22.1	0.9	0.5	22.9	0.8	0.5	15.3	0.7	0.3	21.4
(N = 810)	POL2	0.9	0.6	20.9	1.0	0.6	21.3	0.9	0.7	11.8	0.9	0.5	20.2
	POL3	2.5	1.22	111.7	2.5	1.8	48.9	2.5	1.3	104.6	1.4	0.6	47.5
	RPOL2	0.7	0.4	11.4	0.8	0.4	16.0	0.8	0.6	10.7	0.6	0.3	12.0
	RPOL3	0.7	0.4	42.2	0.9	0.5	28.0	0.8	0.6	18.0	0.6	0.3	38.8
	HPPCC-WCM	0.4	0.1	16.0	0.5	0.2	22.8	0.5	0.4	9.8	0.4	0.1	17.1

The data in Table VII show results for the real imaging setup. Regions of 10 by 10 pixels were taken from each patch and the average used as the RGB value for that region. Each characterisation method was applied using the same training and testing methodology as the simulated experiments.

VII. DISCUSSION

A. Stability and Overtraining

The stability of the methods relates to the ability to resist over-training, and maintain reasonable outputs for unseen testing data.

TABLE IV
CIELAB ΔE ERROR (MEAN, MEDIAN AND MAX) USING LEAVE-ONE-OUT TESTING PARADIGM, WITH RESULTS AVERAGED OVER 31 CAMERAS USING THREE IMAGING ILLUMINANTS. THE HPPCC-WCM USES A p VALUE OF 45.2.

	D65			A			F11			D50		
Method	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
LIN	1.6	0.9	36.4	1.7	0.8	33.6	1.4	0.6	23.9	1.6	0.8	35.1
POL2	1.2	0.7	28.4	1.3	0.7	24.9	1.1	0.6	15.7	1.2	0.7	27.9
POL3	0.9	0.5	22.5	1.0	0.5	20.6	0.9	0.4	11.7	0.9	0.5	22.2
RPOL2	1.0	0.6	28.1	1.1	0.6	29.3	0.9	0.5	24.4	1.0	0.6	26.8
RPOL3	0.8	0.5	24.9	0.9	0.5	23.4	0.7	0.3	12.5	0.8	0.5	24.5
HPPCC-WCM	0.9	0.5	30.2	1.0	0.5	32.4	0.8	0.3	24.8	0.9	0.5	35.7

TABLE V
CIELAB ΔE ERROR (MEAN, MEDIAN AND MAX) FOR A RANGE OF TEST SETS AFTER TRAINING ON MACBETH DC COLORCHECKER (238 PATCHES). RESULTS ARE AVERAGED OVER 31 CAMERAS, WITH 3 DIFFERENT IMAGING ILLUMINANTS. THE HPPCC-WCM USES A p VALUE OF 45.2 FOR EACH CASE.

		D65			A			F11			D50		
Training	Method	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
(N = 238)	MbDC LIN	1.6	0.9	35.5	1.7	0.8	32.0	1.3	0.6	23.4	1.6	0.8	34.5
	POL2	1.2	0.7	28.1	1.2	0.6	24.7	1.0	0.6	15.5	1.1	0.6	27.7
	POL3	0.8	0.4	21.7	0.9	0.5	20.0	0.8	0.4	11.3	0.8	0.4	21.5
	RPOL2	0.9	0.6	25.9	1.0	0.6	24.7	0.9	0.5	21.5	0.9	0.5	26.4
	RPOL3	0.7	0.5	19.3	0.8	0.4	17.4	0.6	0.3	8.4	0.7	0.4	18.9
	HPPCC-WCM	0.8	0.5	19.1	0.8	0.4	22.96	0.7	0.3	12.4	0.8	0.4	19.5
Testing		D65			A			F11			D50		
Esser (N = 264)	LIN	2.9	1.7	45.9	3.2	2.0	57.1	2.5	1.6	35.1	2.9	1.7	45.2
	POL2	2.6	1.4	53.6	3.1	1.6	75.8	2.3	1.3	38.5	2.7	1.4	56.8
	POL3	2.0	1.2	35.6	2.5	1.5	52.5	2.0	1.2	37.2	2.1	1.3	40.8
	RPOL2	2.3	1.1	38.1	2.8	1.3	61.5	2.4	1.0	38.3	2.4	1.2	43.3
	RPOL3	2.7	1.0	106.6	3.2	1.2	137.5	2.9	1.0	93.2	2.8	1.1	115.2
	HPPCC-WCM	1.7	1.0	34.8	1.9	1.2	37.3	1.6	1.0	38.8	1.7	1.1	34.9
Munsell (N = 1269)	LIN	1.4	0.8	29.5	1.8	0.9	26.1	1.4	0.7	18.8	1.5	0.8	28.8
	POL2	1.2	0.9	24.4	1.3	0.8	24.4	1.1	0.7	12.3	1.2	0.8	24.3
	POL3	1.0	0.7	23.5	1.1	0.8	23.1	1.0	0.7	14.3	1.0	0.7	22.8
	RPOL2	1.0	0.6	21.2	1.1	0.7	23.3	1.0	0.6	17.0	1.0	0.6	21.4
	RPOL3	0.9	0.5	17.2	1.0	0.6	16.2	0.9	0.5	11.6	0.9	0.5	16.7
	HPPCC-WCM	1.1	0.6	50.1	1.2	0.7	68.3	1.0	0.6	46.4	1.1	0.6	58.7
Natural (N = 4871)	LIN	1.8	1.2	33.0	1.8	1.2	30.0	1.6	1.1	25.9	1.8	1.1	30.0
	POL2	1.9	1.3	101.4	2.0	1.3	143.6	1.6	1.1	72.5	1.8	1.3	108.0
	POL3	1.9	1.1	688.6	2.1	1.2	526.2	1.8	1.0	255.7	1.8	1.1	638.6
	RPOL2	1.8	1.2	34.6	2.0	1.2	43.9	1.6	1.1	25.8	1.8	1.2	35.1
	RPOL3	1.9	1.2	79.0	1.9	1.2	66.3	1.6	1.1	37.4	1.8	1.1	68.6
	HPPCC-WCM	1.8	1.2	49.8	1.9	1.1	49.5	1.9	1.1	29.8	1.8	1.1	50.8
Hue planes (N = 5715)	LIN	0.8	0.4	35.5	0.9	0.4	32.0	0.7	0.3	23.4	0.8	0.3	34.5
	POL2	0.8	0.5	28.1	0.8	0.5	26.9	0.7	0.4	15.1	0.8	0.5	27.7
	POL3	0.8	0.5	29.7	0.8	0.5	26.2	0.7	0.3	16.4	0.8	0.4	28.7
	RPOL2	0.6	0.4	25.9	0.7	0.4	24.7	0.6	0.4	21.5	0.5	0.2	26.4
	RPOL3	0.5	0.3	19.1	0.6	0.3	14.6	0.5	0.3	10.9	0.4	0.2	18.5
	HPPCC-WCM	0.3	0.1	18.0	0.4	0.1	18.7	0.3	0.1	12.4	0.3	0.1	19.5

An initial idea of stability can be seen in the *LOO* results of Tables II and IV. For the MbCC chart both the RPOL3 and POL3 methods produce worse results than RPOL2 and POL2 respectively, which suggests over-training. In this case, there are 24 colored training patches (with 19 distinct chromaticities) and 19 and 14 degrees of freedom respectively for

the POL3 and RPOL3 methods. For the MbDC data there are more patches to constrain the methods, and the *LOO* results suggest improved stability for the 3^{rd} order methods. However, results for maximum error on the Natural, and Esser charts in Tables III and V show that the 3^{rd} order methods POL3 and RPOL3 can produce very large errors. This may be caused

TABLE VI
INCLUDING NOISE IN THE SIMULATIONS. CIELAB ΔE ERROR (MEAN, MEDIAN AND MAX) ON THE ESSER CHART (TESTING) AFTER TRAINING ON MbDC DATA. THE HPPCC-WCM METHOD USES A p VALUE OF 45.2.

	D65			A			F11			D50		
Method	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
LIN	4.1	3.1	46.6	4.3	3.2	51.8	3.7	2.9	38.0	4.1	3.1	45.2
POL2	3.6	2.6	51.6	3.9	2.7	77.6	3.4	2.5	40.0	3.6	2.7	58.2
POL3	3.2	2.5	38.1	3.4	2.6	52.0	3.1	2.4	38.2	3.2	2.4	37.4
RPOL2	3.3	2.4	37.1	3.6	2.5	57.1	3.4	2.4	37.7	3.3	2.5	40.0
RPOL3	3.7	2.6	63.2	4.0	2.6	115.9	3.7	2.4	70.5	3.7	2.6	122.8
HPPCC-WCM	3.3	2.6	38.7	3.4	2.6	42.1	2.9	2.4	40.1	3.0	2.4	34.6

by the training chart having fewer degrees of freedom than there first appears, since the reflectances are produced by a small number of pigments. It is instructive that this instability is not clear from the *LOO* data, which suggests that higher order polynomials are a risky choice for real situations where the training and test data differ. In contrast both the LIN and HPPCC-WCM methods show similar maximum errors across datasets and illuminants. Given that the HPPCC-WCM method uses many more degrees of freedom than LIN, this suggests that the hue-plane preserving constraint is stable for challenging training and testing sets.

B. Error minimisation

The results for *LOO* testing in Tables II and IV show that in general HPPCC-WCM performs best on the MbCC data, while on MbDC data RPOL3 performs marginally better than HPPCC-WCM, which in turn performs better than RPOL2.

Results for separate training and test sets in Tables III and Table V show a strong data dependence. Performance for all methods generally improves using the MbDC chart. Based on performance HPPCC-WCM, RPOL2 and RPOL3 are the only practically viable solutions. On MbCC data RPOL3 is unstable leaving RPOL2 and HPPCC-WCM. Here WCM has a clear advantage on training data and hueplanes, RPOL2 and HPPCC-WCM perform on par on Natural data, but RPOL2 performs slightly better than HPPCC-WCM on Esser and Munsell.

On MbDC data RPOL3 generally and HPPCC-WCM especially outperform RPOL2, but RPOL3 still tends to be unstable on Esser data with high maximum and average values. Ignoring this stability issue, RPOL3 performs on par with HPPCC-WCM on training and natural data and has an advantage on the maximum deviations on Munsell data. On hueplanes and Esser data, HPPCC-WCM has a clear advantage, which is also the case on Esser with noise data shown in Table VI.

Ignoring the results where overtraining occurs i.e. POL3 trained and tested on MbCC, WCM is generally the best method, except on MbCC data, where RPOL2 is marginally better on Munsell and Esser.

There are two main reasons for the few cases where other methods outperform HPPCC-WCM. Firstly, HPPCC-WCM is constrained to prioritize hue-plane preservation when optimized. The other methods emphasize the minimization of error at the expense of hue-plane preservation. It is thus remarkable

that the HPPCC-WCM method still generally outperforms the other methods on test data where no hueplanes are present. Secondly, it has been shown that the error minimization component of characterization actually implicitly models the underlying reflectance of the training data [34]. As a result, when the training and testing data are taken from similar reflectance distributions, this component of other methods is likely to lead to better generalization. We see this in the case where we train on the Macbeth ColorChecker and test on the Munsell reflectances, which are created with the same pigments.

In all, counting number of best performances of the 4 best performing methods, excluding the real data where HPPCC-WCM performs better than RPOL2 by the mean value, HPPCC-WCM has 73 (54 excluding hueplane results), RPOL3 40 (37), RPOL2 42 (39) and POL3 28 (28). A few of these best performances are shared across methods.

Based on noise susceptibility, stability across training and testing data and general performance, the WCM method is the transformation method of choice.

C. Hue-plane preservation

The plots in Figure 3 show that target xy values for hue-planes fall on straight lines, whereas the root-polynomial and polynomial methods introduce curvature - violating hue-plane preservation. The RPOL2 and RPOL3 methods impose the same curvature across different luminances leaving hueplanes producing hueplanes as curved lines in chromaticity space. The POL2 and POL3 methods project hueplanes to full 2-dimensional manifolds. While both the LIN and HPPCC-WCM methods preserve hue-planes, LIN introduces more error in the transform. These demonstrations support the theory in Section III and error statistics in Tables III and V, in suggesting that the HPPCC-WCM method is more applicable to real-life color imaging applications where color accuracy is required for 3D objects.

D. Noise and Real Data

The results in Table VI show that the method is stable in the presence of noise, with a reduction in performance that parallels other methods. The mean error on the Esser chart is still lower for HPPCC-WCM than for other methods.

Comparing these results with real data in Table VII shows that the median and max errors are of the same order of

magnitude, although the mean errors are higher in the real data than the simulations with noise, which may be caused by the fact that the real data is trained on MbCC data and the noise data is trained on MbDC data. The training and testing patches in the real setup correspond approximately to Munsell surfaces, and the relative performance of the methods is similar to those for Munsell data imaged under the Tungsten illuminant shown in Table III, with the new HPPCC-WCM method giving the lowest median error.

VIII. CONCLUSION AND FUTURE WORK

This paper proposes a novel color camera characterisation termed HPPCC-WCM with the aim of increasing the number of degrees of freedom available to the transformation, while constraining the transformation to hue-plane preservation. Taken together, the numerical results, demonstrations, and theoretical arguments show that the HPPCC-WCM method combines the stability of the low order linear method with the significantly higher accuracy of the best stable high order polynomial methods whilst retaining outstanding accuracy and stability in the saturated colors. The HPPCC-WCM thus outperforms the existing state of the art on challenging reflectance datasets.

In future work we are considering how to optimize the method for as low a number of training patches as possible. This will also reduce the number of precalculated matrices. Furthermore, being fundamentally a per hue-angle linear estimation method, it may be possible to invert the approach to recover the spectral sensitivity of a camera from its responses to a set of training patches.

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